

Distributed Systems

Implementation of a Cloud-Based File Storage System Using Distributed Server Architecture

Summary

Introduction

Cloud computing relies heavily on distributed systems to provide high availability, fault tolerance, and scalability. As organizations increasingly store sensitive data in the cloud, distributed architectures are used to ensure reliable access to files across multiple servers. This research examines how distributed server architecture improves the performance and reliability of cloud-based systems, especially in large-scale environments.

Goal of the Research

The primary goal of the research is to evaluate the effectiveness of distributed cloud architectures in handling data storage and security challenges. The study aims to analyze how distributed systems enhance fault tolerance, load balancing, and intrusion detection within cloud environments.

Methodology

The researchers conducted an experimental study using a distributed cloud environment supported by multiple servers. Machine learning-based intrusion detection techniques were integrated into the distributed system to monitor traffic and detect abnormal behavior. Various algorithms were trained and tested using standard cloud intrusion datasets, and their performance was evaluated using metrics such as accuracy, detection rate, and false-positive rate.

Results

The results indicate that distributed architectures significantly improve system reliability and resilience against failures. The integration of machine learning techniques enhanced the detection of unauthorized activities within the distributed environment. Neural network-based models achieved higher accuracy in detecting complex intrusion patterns compared to traditional methods, while simpler models performed efficiently under low-complexity scenarios.

Conclusion

The study concludes that distributed server architecture is essential for building reliable and secure cloud-based file storage systems. When combined with intelligent monitoring techniques, distributed systems can effectively manage large volumes of data while maintaining security and performance. These findings are highly relevant for designing departmental cloud storage systems that require continuous availability and protection against unauthorized access.

APA References

Machine learning for intrusion detection in cloud environments: A comparative study.
(2024). *ResearchGate*.

https://www.researchgate.net/publication/387076339_Machine_Learning_for_Intrusion_Detection_in_Cloud_Environments_A_Comparative_Study